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Lab 7(B) - Mathematical and Statistical Filters

In [1]:
```
import cv2
import numpy as np

from IPython.display import Image, display
from matplotlib import pyplot as plt

plt.rc('text', usetex=False)
```

In [2]:
```
# ========== ( Adding nosie to an Image ) =================

# 0. Read an Image
imgPath = 'noise.jpg'
img = cv2.imread(imgPath, 0)

# 1. Function that handle all noise algorithm (Guassian, Exp, Uniform & S
def add_noise(image, noise_type):
    row, col = image.shape

    if noise_type == "gaussian":
        mean = 0
        var = 0.01  # Adjust for more/less noise
        sigma = var**0.5
        gauss = np.random.normal(mean, sigma, (row, col))
        noisy = image / 255 + gauss
        return np.clip(noisy * 255, 0, 255).astype(np.uint8)

    elif noise_type == "salt_pepper":
        s_vs_p = 0.5
        amount = 0.04
        noisy = np.copy(image)
        # Salt (white pixels)
        num_salt = np.ceil(amount * image.size * s_vs_p)
        coords = [np.random.randint(0, i - 1, int(num_salt)) for i in ima
        noisy[tuple(coords)] = 255
        # Pepper (black pixels)
        num_pepper = np.ceil(amount * image.size * (1.0 - s_vs_p))
        coords = [np.random.randint(0, i - 1, int(num_pepper)) for i in i
        noisy[tuple(coords)] = 0
        return noisy
```

```
    elif noise_type == "exponential":
        # Scale is the inverse of the rate (lambda)
        expo = np.random.exponential(scale=10.0, size=(row, col))
        noisy = image + expo
        return np.clip(noisy, 0, 255).astype(np.uint8)

    elif noise_type == "uniform":
        low = -20
        high = 20
        uni = np.random.uniform(low, high, (row, col))
        noisy = image + uni
        return np.clip(noisy, 0, 255).astype(np.uint8)



# 2. Load image in grayscale
img = cv2.imread(imgPath, 0)

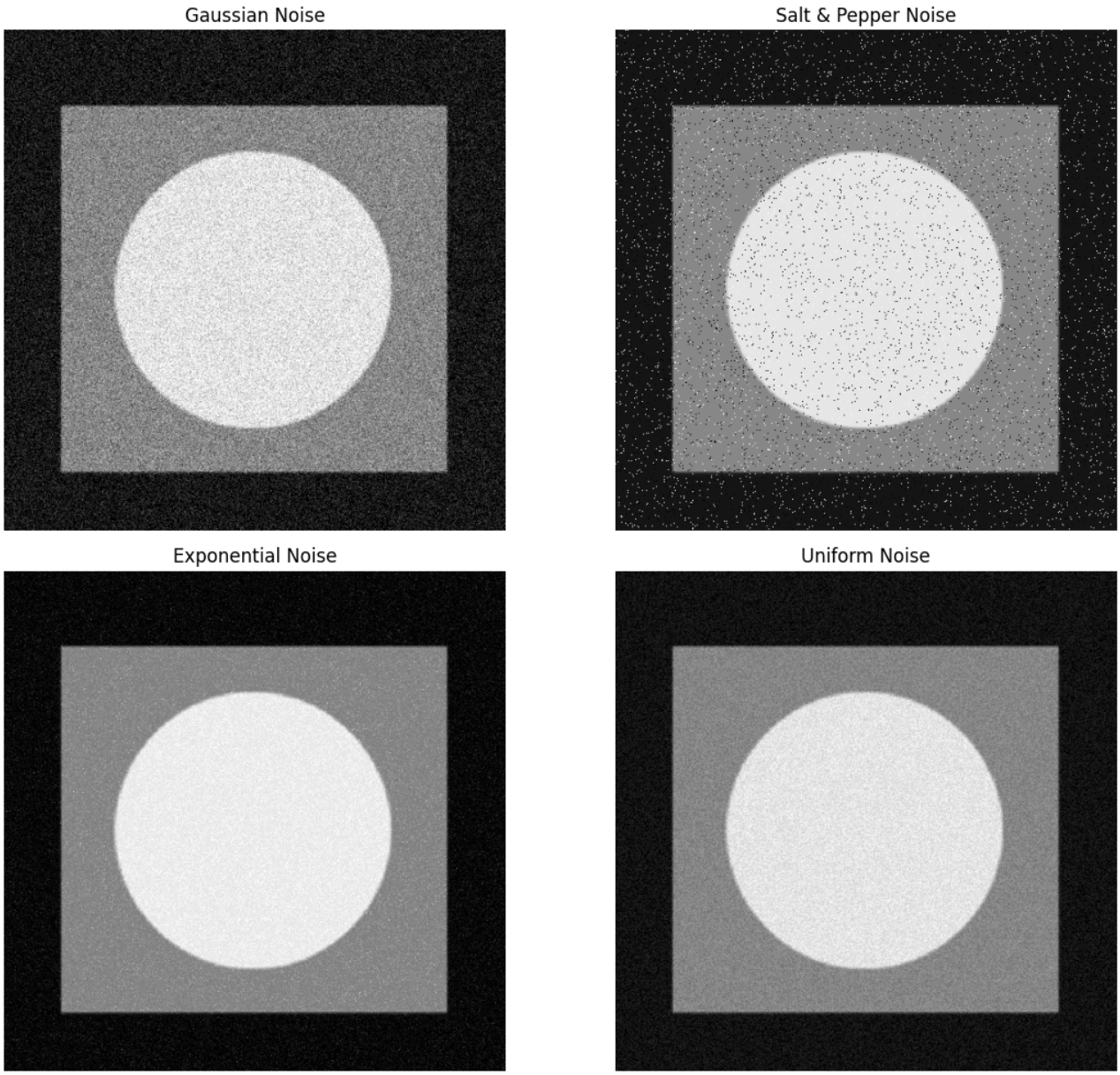
# 3. Generate Noisy Images
noise_gauss = add_noise(img, "gaussian")
noise_sp = add_noise(img, "salt_pepper")
noise_expo = add_noise(img, "exponential")
noise_uni = add_noise(img, "uniform")

# 4. Display in 2x2 Grid
plt.figure(figsize=(12, 10))

plt.subplot(221), plt.imshow(noise_gauss, cmap='gray'), plt.title('Gaussi
plt.subplot(222), plt.imshow(noise_sp, cmap='gray'), plt.title('Salt & Pe
plt.subplot(223), plt.imshow(noise_expo, cmap='gray'), plt.title('Exponen
plt.subplot(224), plt.imshow(noise_uni, cmap='gray'), plt.title('Uniform

for i in range(1, 5):
    plt.subplot(220 + i).axis('off')

plt.tight_layout()
plt.show()
```



In [3]:
```
# --- METHOD 1: Using Built-in OpenCV function ---

# 2. Apply Arithmetic Mean Filter (using a 5x5 blur window)
# The larger the numbers in (3, 3), the smoother the image becomes.,
# you can change kernel size to (5,5), (7,7) etc.
filtered_img = cv2.blur(noise_gauss, (3, 3))

display(filtered_img)
```

```
array([[31, 33, 31, ..., 23, 17, 15],
       [18, 21, 21, ..., 22, 18, 16],
       [26, 22, 18, ..., 17, 18, 18],
       ...,
       [30, 27, 25, ..., 18, 18, 14],
       [23, 26, 17, ..., 27, 26, 20],
       [21, 25, 14, ..., 29, 21, 11]], shape=(506, 508), dtype=uint8)
```

In [4]:
```
# --- METHOD 2: Manual implementation using a Kernel ---

# An Arithmetic Mean kernel is just a matrix of 1s divided by the total n
kernel_size = 3
kernel = np.ones((kernel_size, kernel_size), np.float32) / (kernel_size *
arithmetic_mean_manual = cv2.filter2D(img, -1, kernel)

display(arithmetic_mean_manual)
```

```
array([[24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24],
       ...,
       [24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24],
       [24, 24, 24, ..., 24, 24, 24]], shape=(506, 508), dtype=uint8)
```

Geometric Means

In [5]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

def geometric_mean_filter(image, kernel_size=3):
    # 1. Convert to float64 and add a tiny epsilon to avoid log(0)
    img_float = image.astype(np.float64) + 1e-6

    # 2. Apply the log-transform
    log_img = np.log(img_float)

    # 3. Apply a box filter to the log image (calculates the average of l
    kernel = np.ones((kernel_size, kernel_size), np.float32) / (kernel_si
    mean_log = cv2.filter2D(log_img, -1, kernel)

    # 4. Take the exponent to get back to the geometric mean
    geo_mean = np.exp(mean_log)

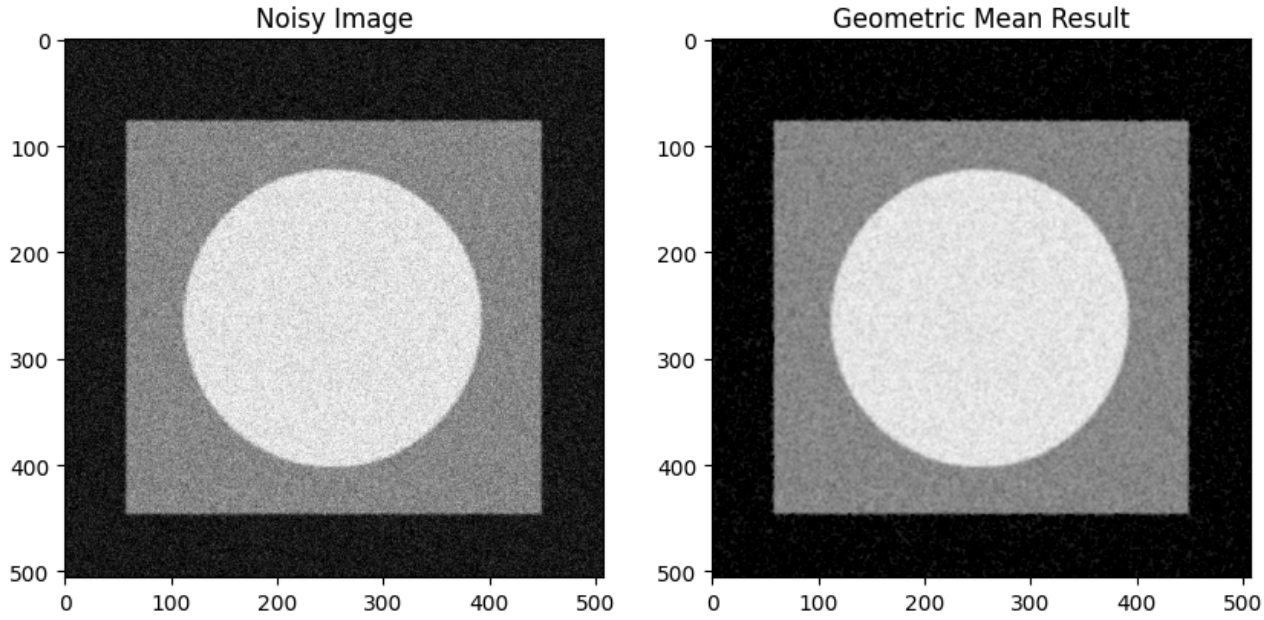
    # 5. Clip and convert back to 8-bit
    return np.clip(geo_mean, 0, 255).astype(np.uint8)


# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# Apply filter
result = geometric_mean_filter(noise_img, kernel_size=3)
```

```
# Display Comparison
plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Noisy Im
plt.subplot(122), plt.imshow(result, cmap='gray'), plt.title('Geometric M
plt.show()
```



Harmonic Means Filter

In [6]:
```
def harmonic_mean_filter(image, kernel_size=3):
    # 1. Convert to float64 to avoid overflow during calculation
    # Add a tiny epsilon to handle black pixels (0)
    img_float = image.astype(np.float64) + 1e-6

    # 2. Calculate the reciprocal of every pixel (1/P)
    reciprocal_img = 1.0 / img_float

    # 3. Sum the reciprocals in a window and divide by window size
    # This is equivalent to applying a standard mean filter to the recipr
    kernel = np.ones((kernel_size, kernel_size), np.float32)
    sum_reciprocal = cv2.filter2D(reciprocal_img, -1, kernel)

    # 4. Final step of the formula: (Window Size) / Sum of Reciprocals
    # Note: Window size is kernel_size * kernel_size
    n = kernel_size * kernel_size
    harmonic_mean = n / sum_reciprocal

    # 5. Clip the results and convert back to 8-bit
    return np.clip(harmonic_mean, 0, 255).astype(np.uint8)

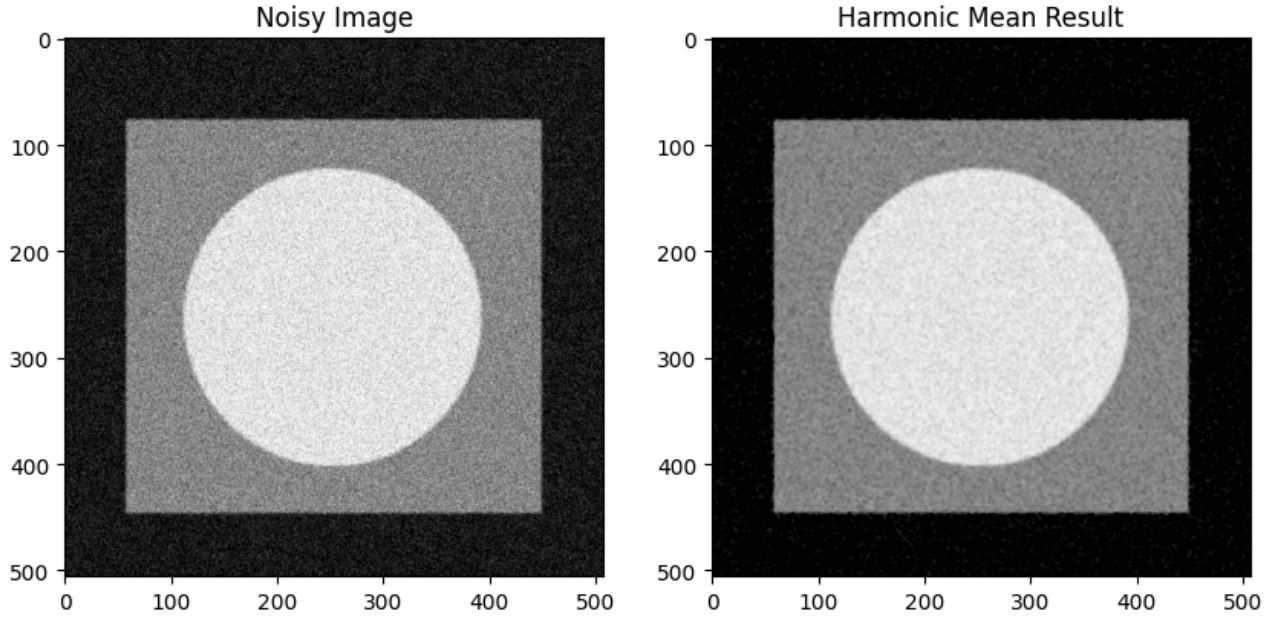

# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
```

```
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# Apply the filter
cleaned_img = harmonic_mean_filter(noise_img, kernel_size=3)

# Display Comparison
plt.figure(figsize=(10, 5))
plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Noisy Im
plt.subplot(122), plt.imshow(cleaned_img, cmap='gray'), plt.title('Harmon
plt.show()
```



Contra - Harmonic Mean Filter

In [7]:
```
def contra_harmonic_mean(image, kernel_size=3, Q=0):
    # 1. Convert to float to handle high powers and prevent overflow
    img_float = image.astype(np.float64) + 1e-6

    # 2. Calculate numerator: Sum of pixels to the power (Q + 1)
    numerator = np.power(img_float, Q + 1)
    kernel = np.ones((kernel_size, kernel_size), np.float32)
    num_sum = cv2.filter2D(numerator, -1, kernel)

    # 3. Calculate denominator: Sum of pixels to the power (Q)
    denominator = np.power(img_float, Q)
    den_sum = cv2.filter2D(denominator, -1, kernel)

    # 4. Divide and clean up
    # Handle potential division by zero
```

```
    result = num_sum / (den_sum + 1e-6)

    return np.clip(result, 0, 255).astype(np.uint8)

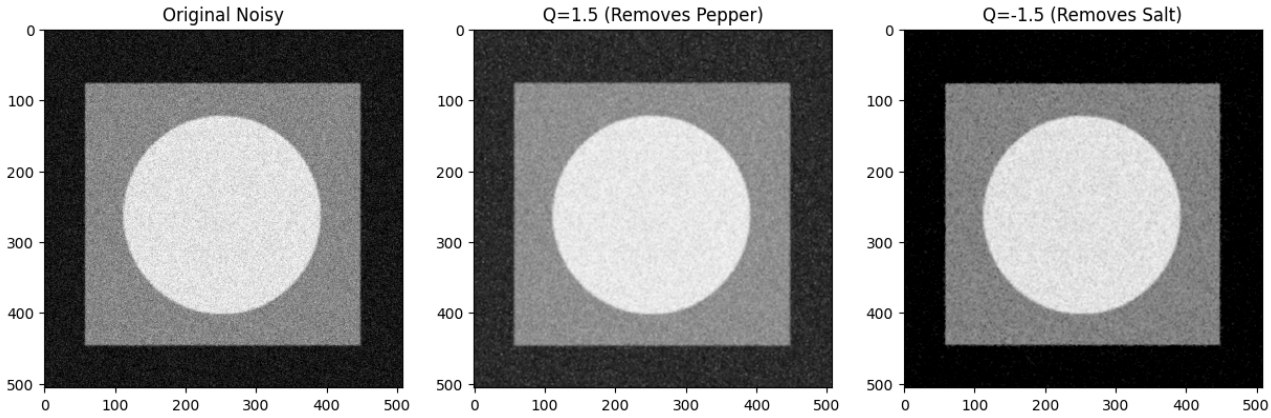
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss

# --- APPLICATIONS ---
# Q > 0 (e.g., 1.5): Removes Pepper Noise (black spots)
pepper_removed = contra_harmonic_mean(noise_img, kernel_size=3, Q=1.5)

# Q < 0 (e.g., -1.5): Removes Salt Noise (white spots)
salt_removed = contra_harmonic_mean(noise_img, kernel_size=3, Q=-1.5)

# --- DISPLAY ---
plt.figure(figsize=(15, 5))
plt.subplot(131), plt.imshow(noise_img, cmap='gray'), plt.title('Original
plt.subplot(132), plt.imshow(pepper_removed, cmap='gray'), plt.title('Q=1
plt.subplot(133), plt.imshow(salt_removed, cmap='gray'), plt.title('Q=-1.
plt.show()
```



Median Filter

In [8]:

```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

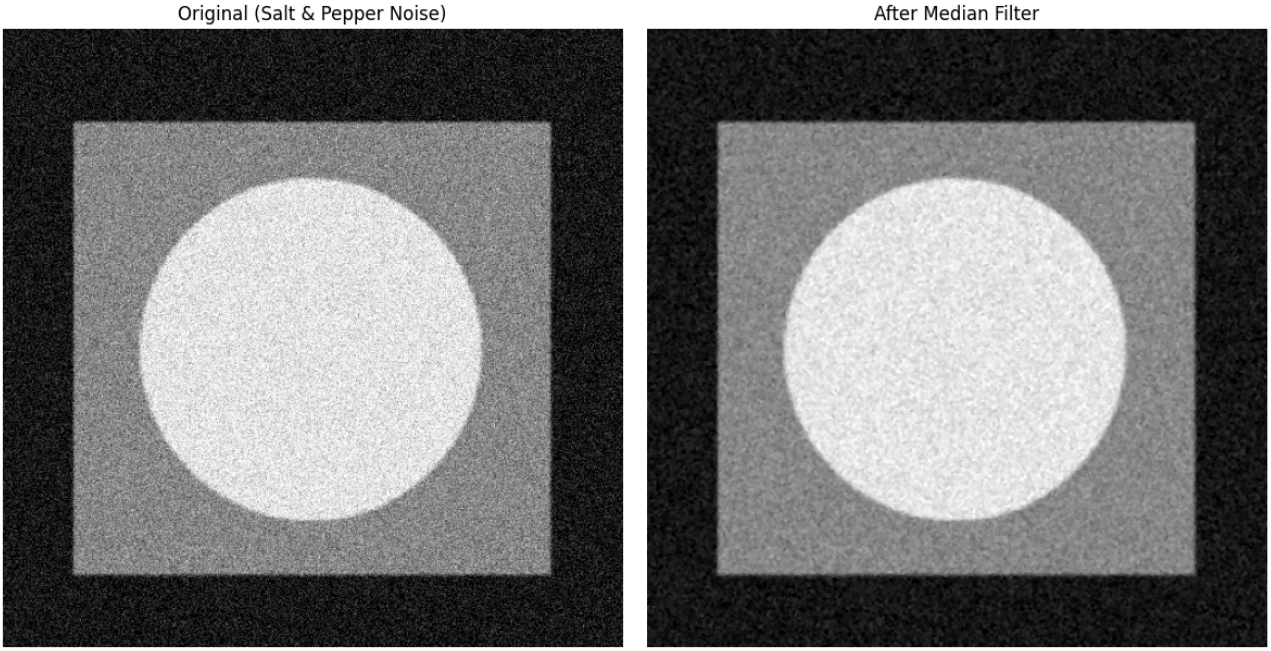
noise_img = noise_gauss

# 2. Apply Median Filter
# The second argument is the 'ksize' (kernel size).
# It MUST be an odd integer (3, 5, 7, etc.)
```

```
median_result = cv2.medianBlur(noise_img, 3)

# 3. Compare Results
plt.figure(figsize=(12, 6))

plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Original
plt.subplot(122), plt.imshow(median_result, cmap='gray'), plt.title('Afte
plt.tight_layout()
plt.show()
```



Max and Min Filter

In [9]:
```
# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")

noise_img = noise_gauss



# 2. Define the Kernel size (e.g., 3x3)
kernel = np.ones((3, 3), np.uint8)

# 3. Apply Max Filter (Dilation)
# This picks the brightest pixel in the neighborhood.
max_filter = cv2.dilate(noise_img, kernel)

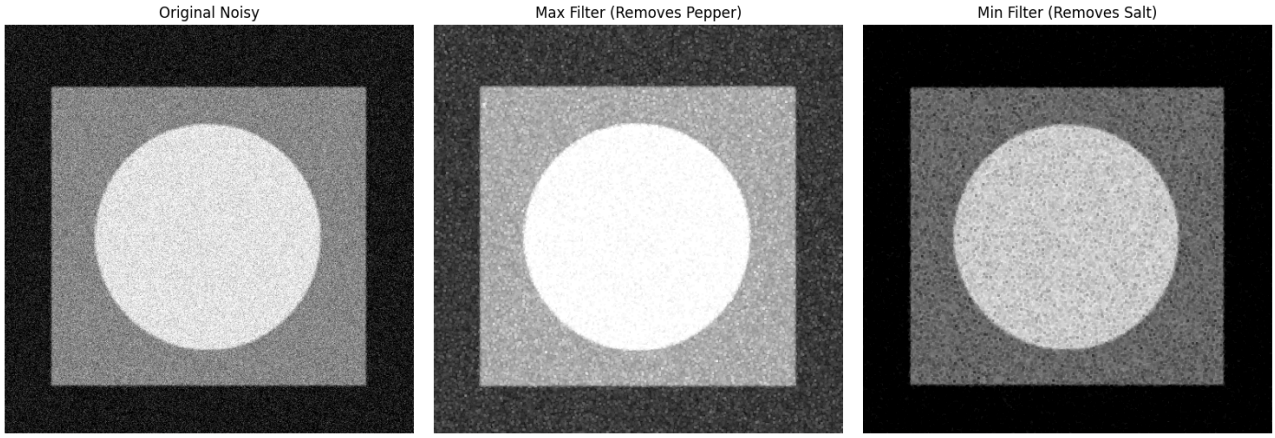
# 4. Apply Min Filter (Erosion)
# This picks the darkest pixel in the neighborhood.
```

```
min_filter = cv2.erode(noise_img, kernel)

# --- Display Results ---
plt.figure(figsize=(15, 5))

plt.subplot(131), plt.imshow(noise_img, cmap='gray'), plt.title('Original
plt.subplot(132), plt.imshow(max_filter, cmap='gray'), plt.title('Max Fil
plt.subplot(133), plt.imshow(min_filter, cmap='gray'), plt.title('Min Fil

for ax in plt.gcf().axes: ax.axis('off')
plt.tight_layout()
plt.show()
```



Mid-Point Filtering

In [10…

```
def midpoint_filter(image, kernel_size=3):
    # 1. Define the structural element (kernel)
    kernel = np.ones((kernel_size, kernel_size), np.uint8)

    # 2. Find the Maximum value in the neighborhood (Dilation)
    max_img = cv2.dilate(image, kernel).astype(np.float64)

    # 3. Find the Minimum value in the neighborhood (Erosion)
    min_img = cv2.erode(image, kernel).astype(np.float64)

    # 4. Calculate the average of Max and Min
    midpoint = (max_img + min_img) / 2

    # 5. Convert back to 8-bit
    return midpoint.astype(np.uint8)



# === List of noisy images
# noise_gauss = add_noise(img, "gaussian")
# noise_sp = add_noise(img, "salt_pepper")
# noise_expo = add_noise(img, "exponential")
# noise_uni = add_noise(img, "uniform")
```

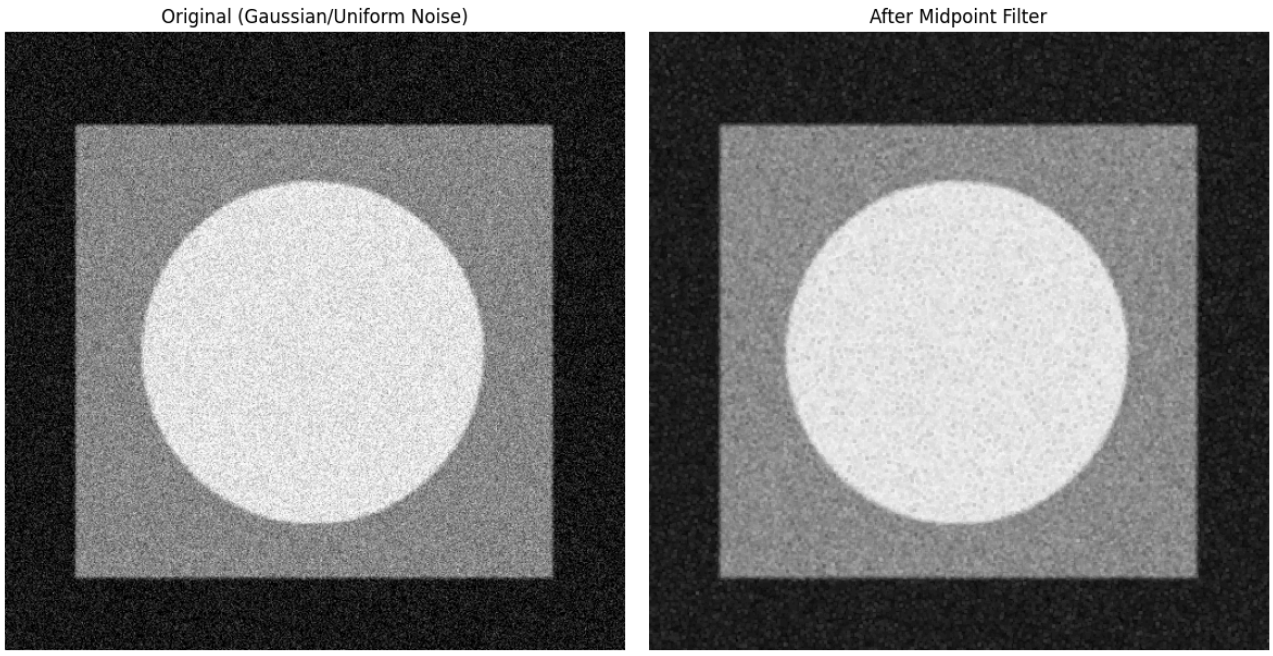
```
noise_img = noise_gauss


# Apply the Midpoint Filter
result = midpoint_filter(noise_img, kernel_size=3)

# --- Display Results ---
plt.figure(figsize=(12, 6))

plt.subplot(121), plt.imshow(noise_img, cmap='gray'), plt.title('Original
plt.subplot(122), plt.imshow(result, cmap='gray'), plt.title('After Midpo

plt.tight_layout()
plt.show()
```



Removing Salt & Pepper Noise from `salt_Pepper.jpg` and `salt_Pepper2.jpg`

We compare the filters above on the two provided salt & pepper images.

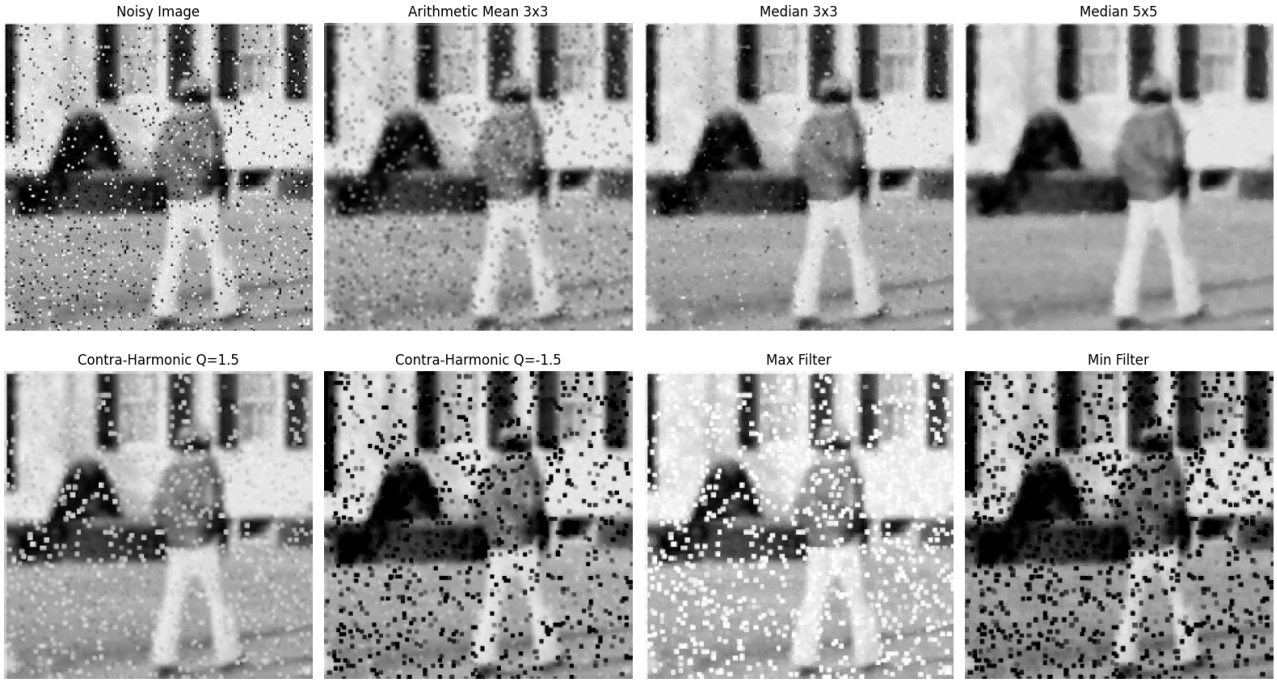
In [11…

```
# ---- Image 1: salt_Pepper.jpg ----
sp1 = cv2.imread('salt_Pepper.jpg', 0)
k3 = np.ones((3, 3), np.uint8)

results1 = {
    'Noisy Image'             : sp1,
    'Arithmetic Mean 3x3'     : cv2.blur(sp1, (3, 3)),
```

```
    'Median 3x3'               : cv2.medianBlur(sp1, 3),
    'Median 5x5'               : cv2.medianBlur(sp1, 5),
    'Contra-Harmonic Q=1.5'    : contra_harmonic_mean(sp1, 3, 1.5),
    'Contra-Harmonic Q=-1.5'   : contra_harmonic_mean(sp1, 3, -1.5),
    'Max Filter'               : cv2.dilate(sp1, k3),
    'Min Filter'               : cv2.erode(sp1, k3),
}

plt.figure(figsize=(16, 9))
for i, (title, im) in enumerate(results1.items(), 1):
    plt.subplot(2, 4, i), plt.imshow(im, cmap='gray', vmin=0, vmax=255),
plt.tight_layout()
plt.show()
```



Observation (image 1): this image has both salt (white) and pepper (black) noise. The arithmetic mean only smears the dots. Contra-harmonic with Q > 0 removes pepper but keeps the salt, and Q < 0 does the opposite, Max removes pepper but enlarges salt, Min does the reverse. The **median filter** removes both types at once while keeping edges, so it is the best choice here (3x3 keeps more detail, 5x5 is cleaner but blurrier).

In [12…

```
# ---- Image 2: salt_Pepper2.jpg ----
# The noise dots here are ~6 px blobs (the image was enlarged), so a 3x3
sp2 = cv2.imread('salt_Pepper2.jpg', 0)
h, w = sp2.shape

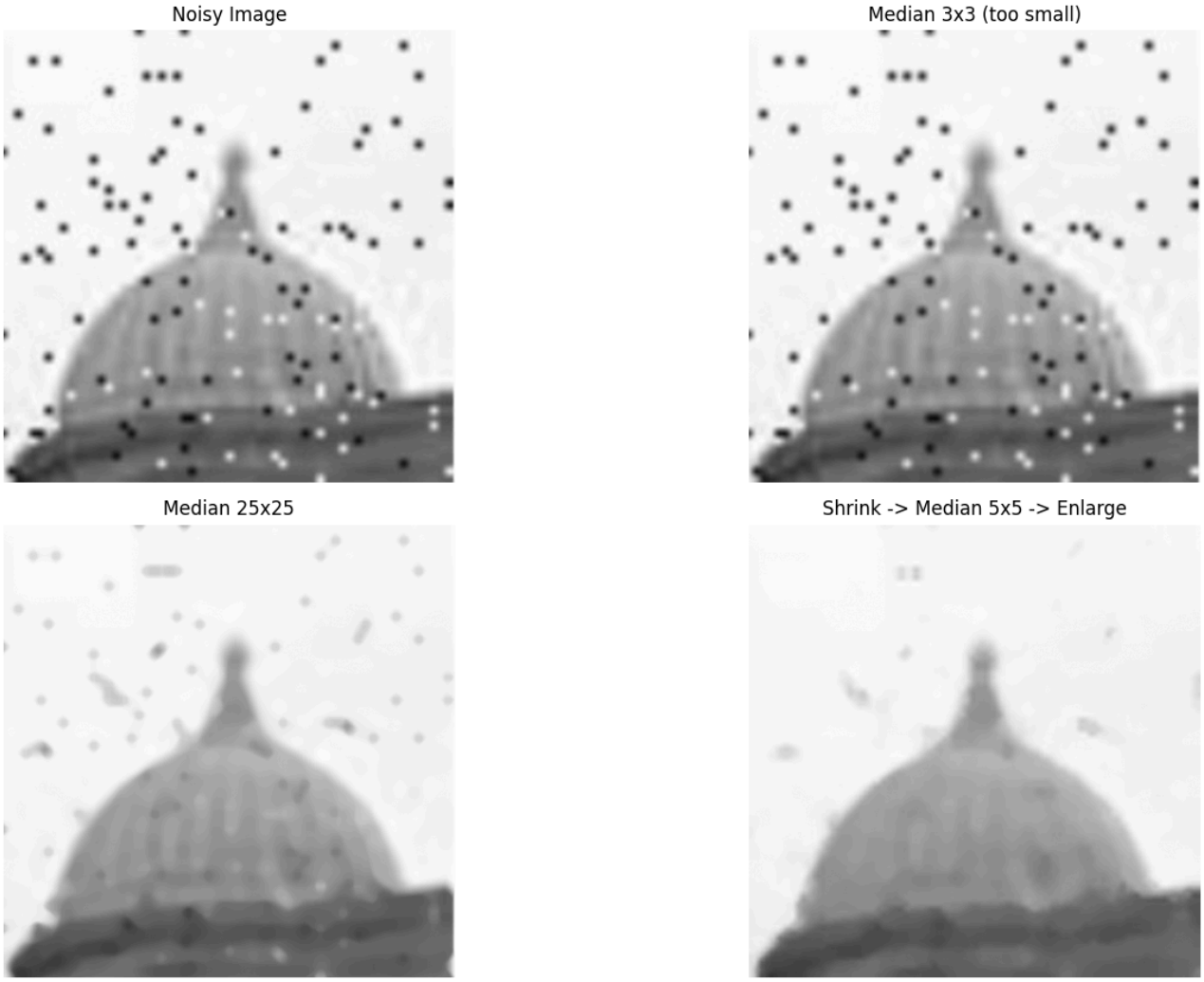
# Option 1: a very large median window (removes the dots but also smears
median25 = cv2.medianBlur(sp2, 25)

# Option 2: shrink so each dot is ~1 pixel, apply a small median, then en
```

```python
f = 7                                                          # approx. enl
small = cv2.resize(sp2, (w // f, h // f), interpolation=cv2.INTER_NEAREST
small_med = cv2.medianBlur(small, 5)
restored = cv2.resize(small_med, (w, h), interpolation=cv2.INTER_CUBIC)

results2 = {
    'Noisy Image'                          : sp2,
    'Median 3x3 (too small)'               : cv2.medianBlur(sp2, 3),
    'Median 25x25'                         : median25,
    'Shrink -> Median 5x5 -> Enlarge'      : restored,
}

plt.figure(figsize=(16, 9))
for i, (title, im) in enumerate(results2.items(), 1):
    plt.subplot(2, 2, i), plt.imshow(im, cmap='gray', vmin=0, vmax=255),
plt.tight_layout()
plt.show()
```



Observation (image 2): the noise is mostly pepper (black dots) with some salt (white dots), but each dot is a ~6x6 blob, so a 3x3 median cannot remove

it (the blob fills more than half of the window). A very large median (25x25) removes the dots but leaves faint ghosts and blurs the dome. Shrinking the image so that each dot is about one pixel, applying a 5x5 median, and enlarging back gives the cleanest result while keeping the dome shape.